ANTHONY JARJOUR B.S.

+1 (915) 274-8704
anthonyjarjour10@gmail.com
linkedin.com/in/anthony-jarjour

EDUCATION

California State University, Fullerton California, United States
B.S. Computer Science, Minor in Mathematics and Physics 2021.08 - 2026.05

- Extended degree by one year to formally pursue advanced physics and mathematics coursework and research in theoretical physics.

California State University, Fullerton – Open University California, United States
Post-Baccalaureate Coursework and Research 2026.08 – Present

- Continuing advanced coursework in mathematics and physics in preparation for graduate study while conducting research in Riemannian geometry and theoretical physics.

RESEARCH

Research Assistant | Mathematics – Riemannian geometry 2026.05 – Present

- Investigated positive and nonnegative sectional curvature on biquotient manifolds $SU(n)//S^1$ following Eschenburg (1984) and Ziller (2018), focusing on Cheeger deformations, Riemannian submersions, O'Neill's formula, quotient metrics, and zero-curvature planes.
- Developed an obstruction to strictly positive sectional curvature for Cheeger-deformed $SU(n)//S^1$ metrics with $n \geq 4$ by identifying surviving flat torus directions, extending techniques from positively curved Eschenburg spaces $SU(3)//S^1$.

Research Assistant | Mathematical Physics – QFT 2025.05 – Present

- Investigated thermodynamic properties of inequivalent Rindler vacua in Minkowski spacetime following Lochan et al. (2025), focusing on Bogoliubov transformations, vacuum inequivalence, and Planck-scale aspects of quantum field theory.
- Built graduate-level theoretical foundation in General Relativity and Quantum Field Theory using Schutz (1985), Wald (1994), Ashtekar (1975), and Panangaden (1979), covering algebraic QFT, Green's functions and propagators, and Kähler structures.

PRESENTATION

Curvature and Biquotients | *2026 CSUF Summer Symposium* 2026.08

- Exhibited a poster related to work done as a part of the Math Summer Research Program centered around examples of manifolds with strictly positive sectional curvature, with a novel construction similar to Eschenburg space and detailed work showcasing its failure to admit strictly positive sectional curvature.

Bogoliubov Transformations Between Shifted Rindler Frames | 2026.01
Joint Mathematics Meetings (JMM)

- Presented at the 2026 Joint Mathematics Meetings an expanded study of Bogoliubov transformations between shifted Rindler frames, exploring how different quantization schemes yield distinct vacuum structures and physical interpretations.

Bogoliubov Transformations Between Shifted Rindler Frames | 2025.10
Math Research Symposium

- Delivered an in-depth exposition of Lochan & Padmanabhan (2025), reproducing their derivation of Bogoliubov transformations between shifted Rindler frames and analyzing the resulting inequivalent vacuum structure in detail.

Data Science in Sport Analytics | *ECS Innovation Expo* 2025.04

- Cleaned and preprocessed soccer match data to develop machine learning models for predicting game outcomes, and presented application at an expo.

Awards & Honors

Dean's List | *California State University, Fullerton* 2021.08 - 2026.05

- Recognized for academic excellence for the Fall 2024, Spring 2025, Fall 2025, and Spring 2026 semesters.

Funded MSRP Award | *CSUF Department of Mathematics* 2026.05

- Selected for the CSUF Department of Mathematics Math Summer Research Program to conduct 100 hours of paid undergraduate research.
- Worked under faculty mentorship on a mathematics research project, culminating in a research poster and summary of findings.

3rd Place Award | *CSUF CS Showcase* 2025.04

- Awarded 3rd place for developing a machine learning model to predict soccer match outcomes, presented at the Computer Science Showcase at California State University, Fullerton.

Teaching Experience

Teaching Assistant - CSUF | Fullerton, California Fall 2026

- Created course quizzes aligned with the lecture material and course objectives with answer keys and worked solutions to support grading consistency.
- Graded quizzes and provided consistent evaluation of student work using established criteria.

Mathematical Circle Instructor - CSUF | Fullerton, California Spring 2026

- Lead a problem-solving session for middle- and high-school students in an open university outreach program, delivering a lecture and workshop-style activities.
- Designed and curated challenging problem sets and guided solutions to strengthen mathematical reasoning, proof skills, and creativity.
- Collaborated with faculty mentorship to select topics, refine lesson plans, and adapt instruction to varied student backgrounds and ability levels.

Undergraduate Tutor (Volunteer) - CSUF | Fullerton, California Fall 2025

- Mentored undergraduate students through homework and exam preparation in vector and tensor analysis.
- Broke down complex derivations into approachable steps and emphasized geometric/physical interpretation alongside formalism.
- Developed targeted practice problems and review notes to address recurring conceptual gaps.

Relevant Coursework

Completed

Computer Science: Compilers and Languages; Algorithm Engineering; Theory of Computation; Artificial Intelligence; Machine Learning

Mathematics: Vector and Tensor Analysis; Mathematical Probability; Partial Differential Equations; Differential Geometry; Quantum Field Theory; Intro to General Relativity

Physics: Thermodynamics, Kinetic Theory, and Statistical Physics; Classical Dynamics

Planned

Real Analysis I; Linear Algebra; Abstract Algebra; Topology; Electromagnetic Theory I–II; Quantum Mechanics I–II

Academic Interests

Theoretical Physics, Quantum Field Theory, General Relativity, Quantum Gravity, Computational Physics, Machine Learning for Physical Systems.

Projects

Large Language Model for Code Generation 2024.09 – 2024.11

- Technologies: Python, NLTK, Numpy, PyTorch, Pandas
- Notes: Evaluated model output using unit tests to validate code correctness across multiple functional tasks, with a 93% pass rate.

N-Body Problem Simulator 2024.08 – 2024.10

- Technologies: Python, Numpy, MatPlotLib
- Notes: Implemented Runge-Kutta integration to model two-body dynamics in a Newtonian gravitational system; visualized orbit evolution and field behavior.
- Simulated gravitational fields using Newtonian mechanics with relativistic corrections

Residual Neural Network Image Classifier 2024.01 – 2024.05

- Technologies: Python, PyTorch, NodeJS, Javascript, FastAPI, ScikitLearn, Pandas
- Notes: Implemented an RNN based on K. He, X. Zhang, S. Ren, and J. Sun, ”Deep residual learning for image recognition,” arXiv preprint arXiv:1512.03385, 2015, and achieved 95% validation accuracy with 0.25 loss, on bird species images

Skills

Programming Languages:
Python, C, C++, JavaScript, PHP, HTML, MATLAB, R

Machine Learning / Data Science:
PyTorch, Torchvision, NumPy, Pandas, Scikit-learn, Matplotlib, Hugging Face Hub, Jupyter Notebooks

Web / App Development:
React Native, Expo, Node.js, npm, FastAPI, Uvicorn, REST APIs

Tools / Systems:
Git, GitHub, Linux/Ubuntu, VS Code, SQLite/MySQL-style database design

Internships

Cybersecurity Analyst - El Paso Water Utility | El Paso, Texas 2022.06 - 2022.08

- Designed and automated reports to visualize trends across multidimensional data streams, enhancing system performance and predictive diagnostics.

Selected References

B. Wilking and W. Ziller, “Revisiting Homogeneous Spaces with Positive Curvature,” *Journal f”ur die reine und angewandte Mathematik*, vol. 738, pp. 313–328, 2018.

J.-H. Eschenburg, *Freie isometrische Aktionen auf kompakten Lie-Gruppen mit positiv gekr”ummten Orbitr”aumen*, Schriftenreihe des Mathematischen Instituts der Universit”at M”unster, 2. Serie, vol. 32. Universit”at M”unster, Mathematisches Institut, M”unster, 1984.

K. Lochan and T. Padmanabhan, “A Nested Sequence of Inequivalent Rindler Vacua: Universal Relic Thermality of Planckian Origin,” *Classical and Quantum Gravity*, vol. 42, no. 3, 03LT01, 2025.

B. F. Schutz, *A First Course in General Relativity*. Cambridge University Press, 1985.

R. M. Wald, *Quantum Field Theory in Curved Spacetime and Black Hole Thermodynamics*. University of Chicago Press, 1994.

A. Ashtekar and A. Magnon, “Quantum Fields in Curved Space-Times,” *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, vol. 346, no. 1646, pp. 375–394, 1975.

P. Panangaden, “Positive and Negative Frequency Decompositions in Curved Spacetime,” *Journal of Mathematical Physics*, vol. 20, no. 12, pp. 2506–2510, 1979.